

8.0 RECEIVE FILTERS

To minimize the processing requirements of the host controller, the ENC28J60 incorporates several different receive filters which can automatically reject packets which are not needed. Six different types of packet filters are implemented:

- Unicast
- Pattern Match
- Magic Packet™
- Hash Table
- Multicast
- Broadcast

The individual filters are all configured by the ERXFCON register (Register 8-1). More than one filter can be active at any given time. Additionally, the filters can be configured by the ANDOR bit to either logically AND, or logically OR, the tests of several filters. In other words, the filters may be set so that only packets accepted by all active filters are accepted, or a packet accepted by any one filter is accepted. The flowcharts in Figure 8-1 and Figure 8-2 show the effect that each of the filters will have depending on the setting of ANDOR.

The device can enter Promiscuous mode and receive all packets by clearing the ERXFCON register. The proper setting of the register will depend on the application requirements.

ENC28J60

REGISTER 8-1: ERXFCN: ETHERNET RECEIVE FILTER CONTROL REGISTER

R/W-1	R/W-0	R/W-1	R/W-0	R/W-0	R/W-0	R/W-0	R/W-1
UCEN	ANDOR	CRCEN	PMEN	MPEN	HTEN	MCEN	BCEN
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7 **UCEN:** Unicast Filter Enable bit

When ANDOR = 1:

1 = Packets not having a destination address matching the local MAC address will be discarded

0 = Filter disabled

When ANDOR = 0:

1 = Packets with a destination address matching the local MAC address will be accepted

0 = Filter disabled

bit 6 **ANDOR:** AND/OR Filter Select bit

1 = AND: Packets will be rejected unless all enabled filters accept the packet

0 = OR: Packets will be accepted unless all enabled filters reject the packet

bit 5 **CRCEN:** Post-Filter CRC Check Enable bit

1 = All packets with an invalid CRC will be discarded

0 = The CRC validity will be ignored

bit 4 **PMEN:** Pattern Match Filter Enable bit

When ANDOR = 1:

1 = Packets must meet the Pattern Match criteria or they will be discarded

0 = Filter disabled

When ANDOR = 0:

1 = Packets which meet the Pattern Match criteria will be accepted

0 = Filter disabled

bit 3 **MPEN:** Magic Packet™ Filter Enable bit

When ANDOR = 1:

1 = Packets must be Magic Packets for the local MAC address or they will be discarded

0 = Filter disabled

When ANDOR = 0:

1 = Magic Packets for the local MAC address will be accepted

0 = Filter disabled

bit 2 **HTEN:** Hash Table Filter Enable bit

When ANDOR = 1:

1 = Packets must meet the Hash Table criteria or they will be discarded

0 = Filter disabled

When ANDOR = 0:

1 = Packets which meet the Hash Table criteria will be accepted

0 = Filter disabled

bit 1 **MCEN:** Multicast Filter Enable bit

When ANDOR = 1:

1 = Packets must have the Least Significant bit set in the destination address or they will be discarded

0 = Filter disabled

When ANDOR = 0:

1 = Packets which have the Least Significant bit set in the destination address will be accepted

0 = Filter disabled

bit 0 **BCEN:** Broadcast Filter Enable bit

When ANDOR = 1:

1 = Packets must have a destination address of FF-FF-FF-FF-FF-FF or they will be discarded

0 = Filter disabled

When ANDOR = 0:

1 = Packets which have a destination address of FF-FF-FF-FF-FF-FF will be accepted

0 = Filter disabled

FIGURE 8-1: RECEIVE FILTERING USING OR LOGIC

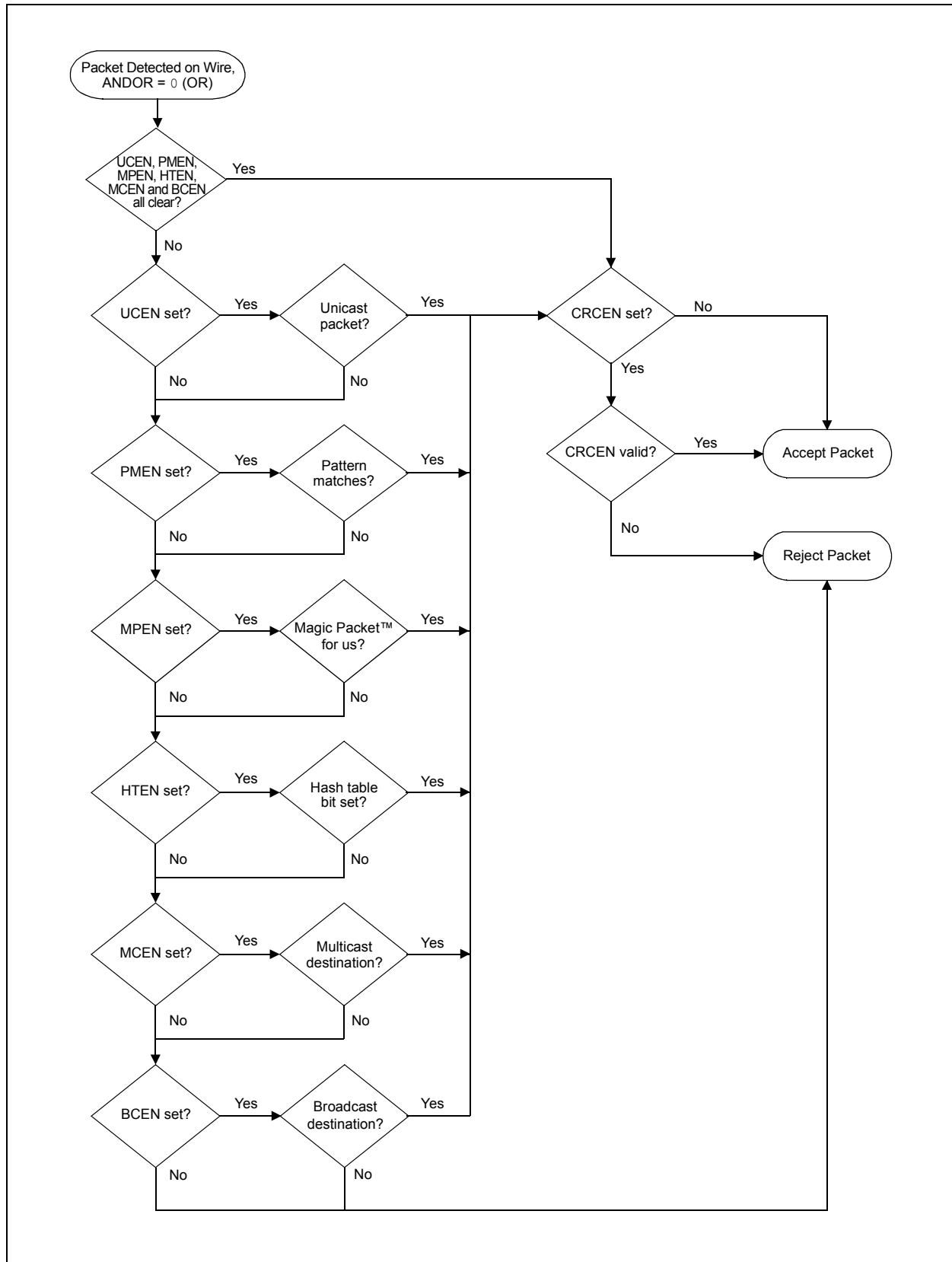
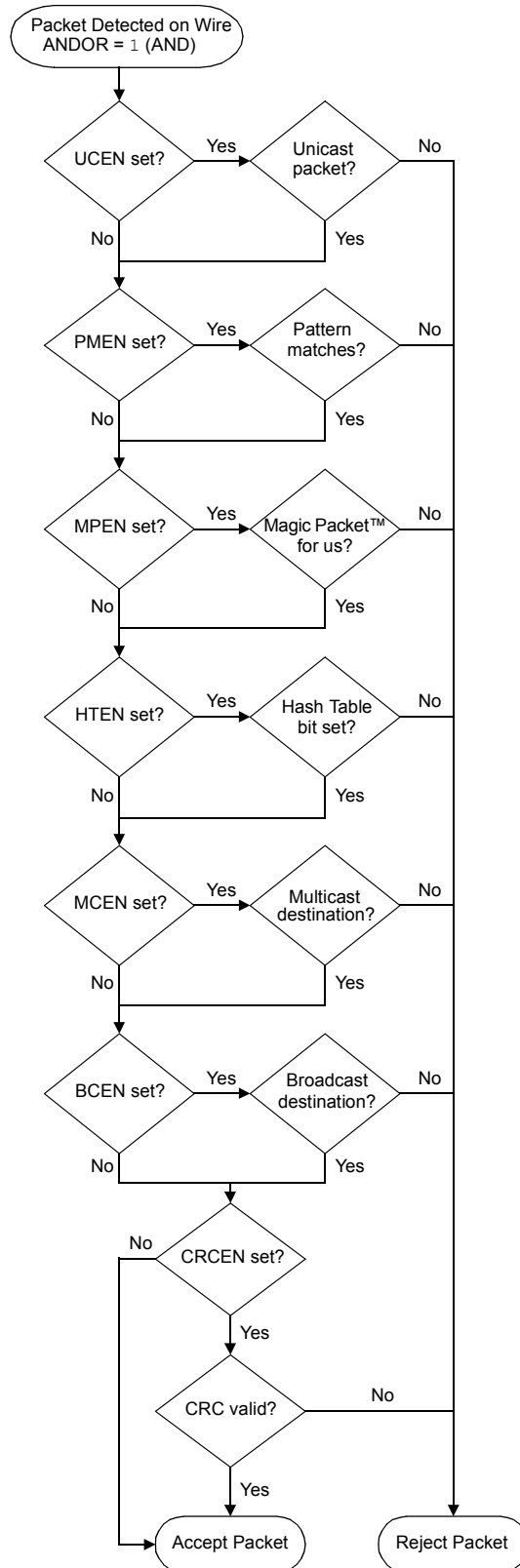


FIGURE 8-2: RECEIVE FILTERING USING AND LOGIC



8.1 Unicast Filter

The Unicast receive filter checks the destination address of all incoming packets. If the destination address exactly matches the contents of the MAADR registers, the packet will meet the Unicast filter criteria.

8.2 Pattern Match Filter

The Pattern Match filter selects up to 64 bytes from the incoming packet and calculates an IP checksum of the bytes. The checksum is then compared to the EPMCS registers. The packet meets the Pattern Match filter criteria if the calculated checksum matches the EPMCS registers. The Pattern Match filter may be useful for filtering packets which have expected data inside them.

To use the Pattern Match filter, the host controller must program the Pattern Match offset (EMOH:EPML), all of the Pattern Match mask bytes (EPMM7:EPMM0) and the Pattern Match Checksum register pair (EPMCSH:EPMCSL). The Pattern Match offset should be loaded with the offset from the beginning of the destination address field to the 64-byte window which will be used for the checksum computation. Within the 64-byte window, each individual byte can be selectively included or excluded from the checksum computation by setting or clearing the respective bit in the Pattern Match mask. If a packet is received which would cause the 64-byte window to extend past the end of the CRC,

the filter criteria will immediately not be met, even if the corresponding mask bits are all '0'. The Pattern Match Checksum registers should be programmed to the checksum which is expected for the selected bytes. The checksum is calculated in the same manner that the DMA module calculates checksums (see **Section 13.2 "Checksum Calculations"**). Data bytes which have corresponding mask bits programmed to '0' are completely removed for purposes of calculating the checksum, as opposed to treating the data bytes as zero.

As an example, if the application wished to filter all packets having a particular source MAC address of 00-04-A3-FF-FF-FF, it could program the Pattern Match offset to 0000h and then set bits 6 and 7 of EPMM0 and bits 0, 1, 2 and 3 of EPMM1 (assuming all other mask bits are '0'). The proper checksum to program into the EPMCS registers would be 0x5BFC. As an alternative configuration, it could program the offset to 0006h and set bits 0, 1, 2, 3, 4 and 5 of EPMM0. The checksum would still be 5BFC. However, the second case would be less desirable as packets less than 70 bytes long could never meet the Pattern Match criteria, even if they would generate the proper checksum given the mask configuration.

Another example of a Pattern Matching filter is illustrated in Figure 8-3.

FIGURE 8-3: SAMPLE PATTERN MATCH FORMAT

